

Session 3

Case Study: Singapore's Smart Nation, Open Government Products, and Singapore Government Tech Stack

Ernest Guevarra

ernest@guevarra.io

University of Oxford



SMART NATION

S I N G A P O R E

Smart Nation journey



1963



1981



1997



2000



2007



2014



2016



2017



2018



2022



2023



2024

Vision

- Singapore's **Smart Nation initiative**, led by the Ministry of Digital Development and Information (MDDI), aims to build a thriving digital future for everyone.
- Launched in **2014**, Smart Nation 1.0 sought to improve lives through seamless and meaningful use of technology.
- Since its launch, Singapore has integrated digital technologies deeply into daily life, creating new opportunities while also needing to manage risks to well-being and social cohesion.
- In **2024**, Singapore introduced **Smart Nation 2.0**, a refreshed vision focused on using technology more effectively to transform the nation's future.
- The new phase shifts from primarily building digital capabilities to applying them strategically to shape Singapore's long-term development.

Goals

Trust



Enhance the security and resilience of our digital infrastructure



Strengthen our fight against harmful online activities



Expand our trusted digital spaces together

Growth



Power our digital economy



Empower our enterprises
and workers



Prepare our next generation

Community



Strengthen inclusion:
leaving no one behind



Strengthen the unity of our
community

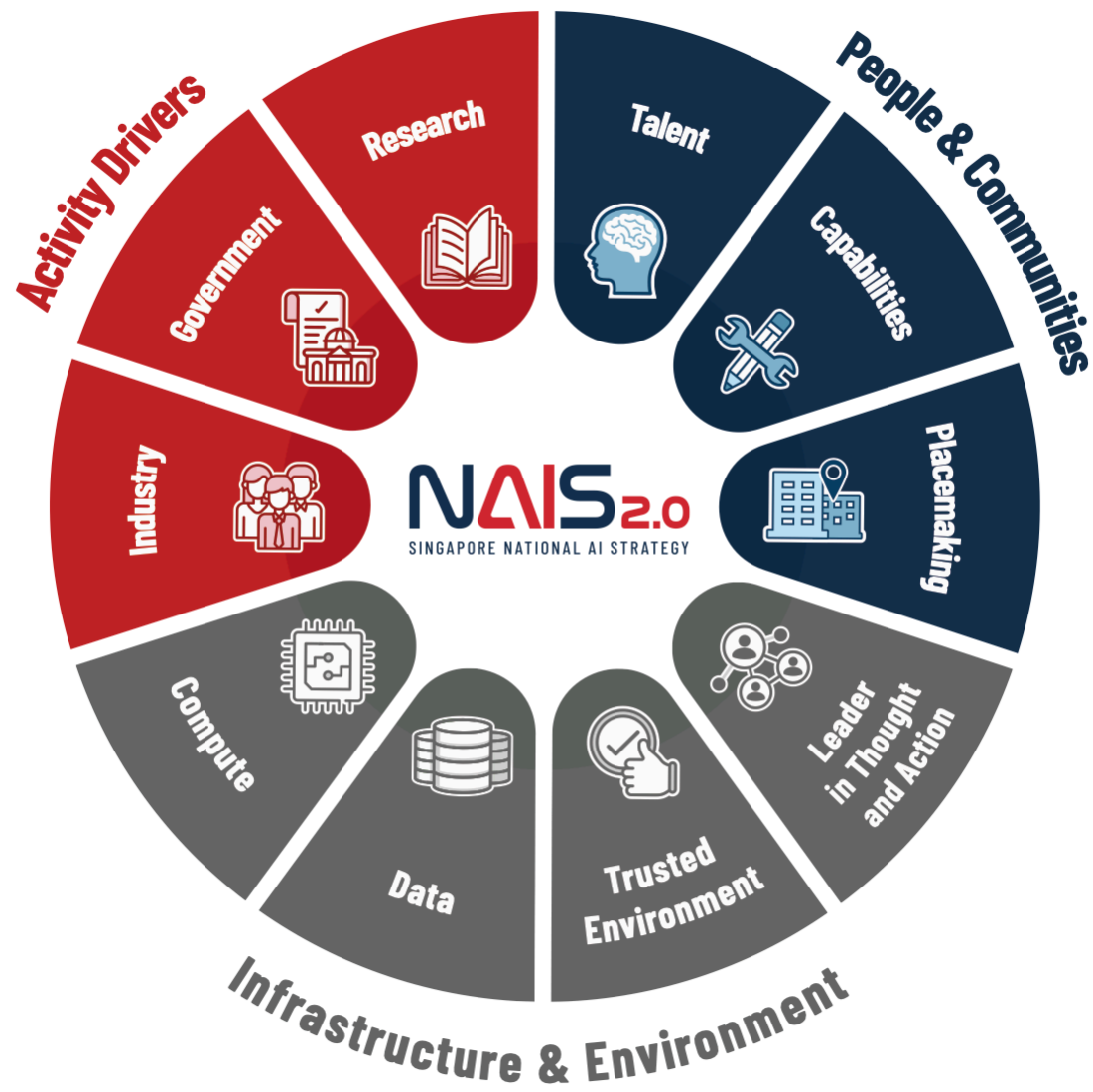


Strengthen partnerships
with citizens

Smart City achievements

- **99%** of all transactions between citizens and the government can be completed online.
- **High citizen and business satisfaction** with Government Digital Services at **83%** in 2023.
- **99%** of resident households are **connected to the internet**, as of 2022.
- **95%** of small and medium enterprises (SMEs) have adopted at least one digital technology from nine categories: cybersecurity, cloud, e-payment, e-commerce, data analytics, artificial intelligence, Internet of Things, blockchain, and immersive media.

National AI Strategy

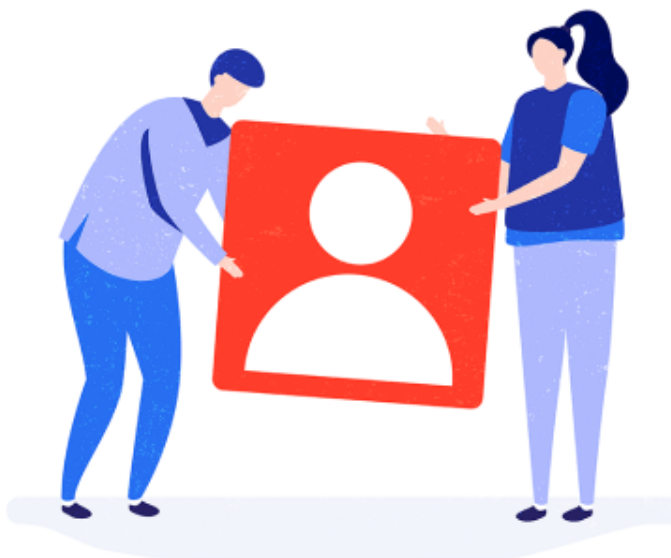


Open Government Products

Approach



Solve real problems



Build for the user



Push for change

Products

- More than 40 products in progress
- Use **report cards** mechanism to provide a transparent overview of the work being done on each of the products

Tackle real problems

Share why and how the product solves a problem with updates along the way.

Resource responsibly

Product costs are made open as possible to stay accountable to taxpayers and citizens.

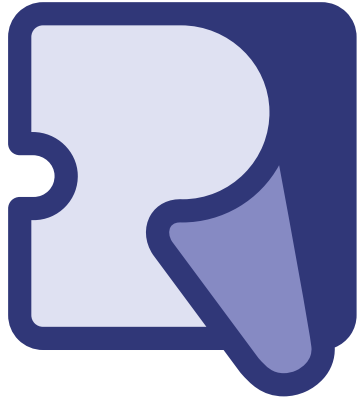
Track impact clearly

Share product outcomes to ensure delivery of clear value to the public.



SCAMSHIELD

- **ScamShield** helps users detect, check, and report suspicious messages, calls, and links.
- Reported scams are verified by authorities, helping them track and respond to new scam variants.



REDEEMSG

- Paper voucher campaigns were slow to launch and difficult for residents to claim and use.
- Merchants experienced long delays in receiving payments.
- Existing digital alternatives were available but scattered across multiple complicated apps.
- **RedeemSG** provides a unified platform for government agencies and non-profits to create, distribute, and monitor vouchers nationwide.
- Citizens redeem vouchers and merchants accept them through the same system.



- Paper-based tracking made it difficult for organisations to monitor distributions or adapt plans in real time.
- This led to long waiting times, inaccurate records, and limited ability to reach more beneficiaries.
- **DistributeSG** enables government agencies and non-profits to run and monitor distribution campaigns with real-time tracking.
- It includes an admin portal for organisers to set up campaigns and a mobile app for on-the-ground tracking of distributed items.

Initiatives



- **Build for Good** is a citizen engagement initiative by Open Government Products that encourages citizens to help improve Singapore.
- People can participate through a **hackathon**, an **accelerator for winning teams**, or by joining the **community**.
- The initiative aims to create a supportive space where individuals feel empowered to contribute positively, experiment, and build solutions that make the world better.



hack for public good

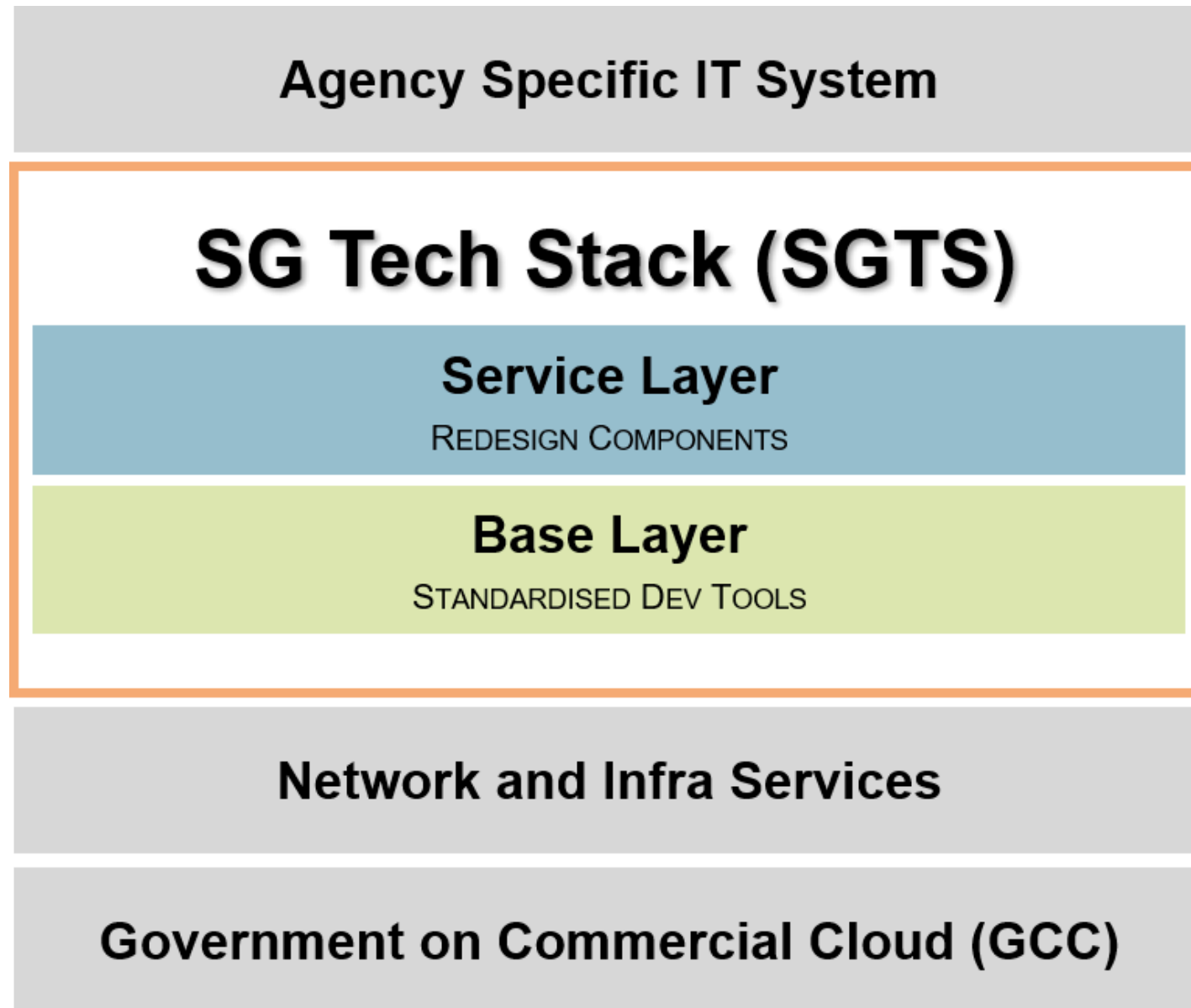
- **Hack for Public Good (HFPG)** replaces top-down product development with a community-driven approach.
- Instead of leadership setting the direction, tech teams engage directly with the community to identify public good problems in Singapore.
- These teams then design and build solutions based on what they discover.

Singapore government tech stack

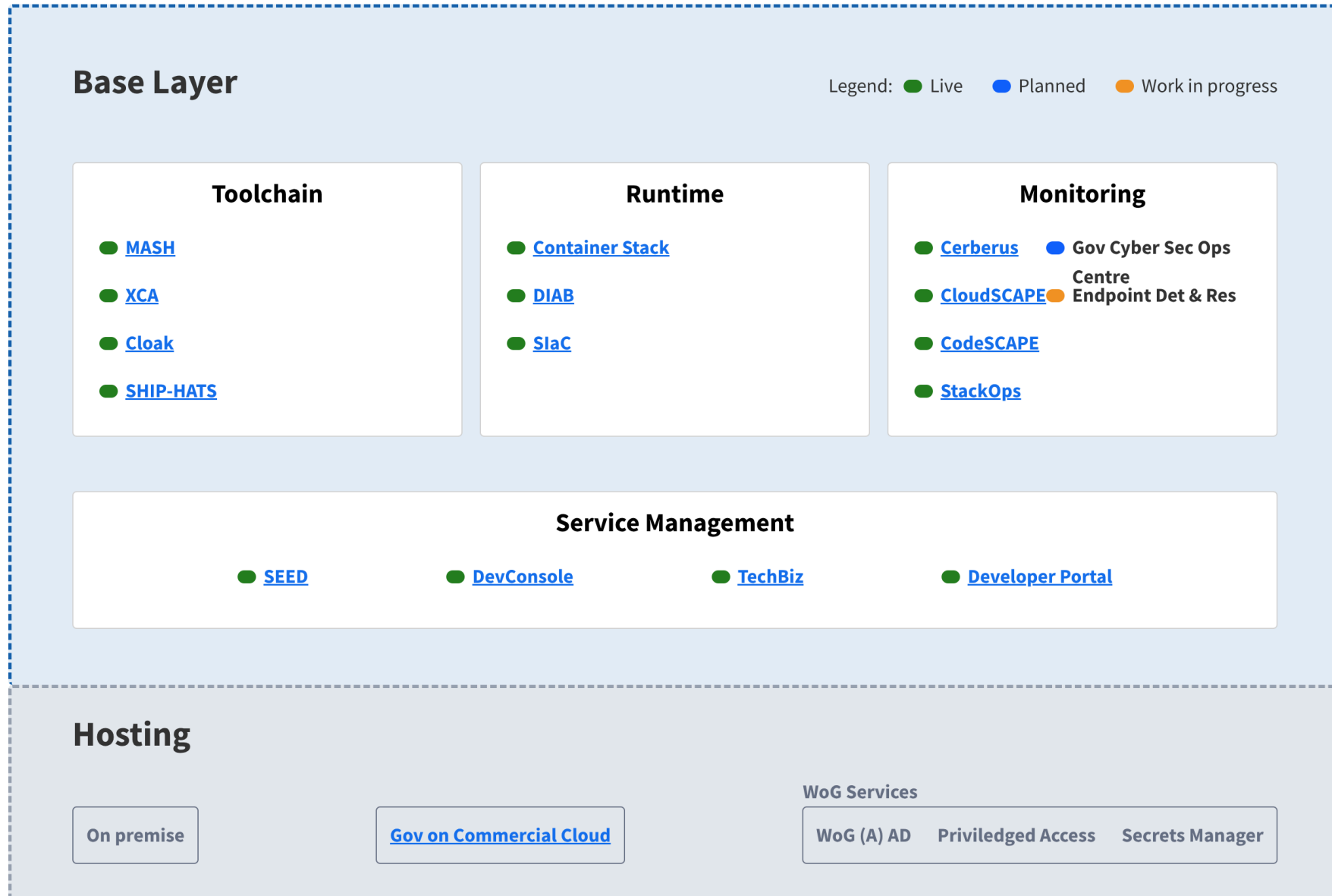
Overview

- The **Singapore Government Tech Stack (SGTS)** modernises development practices across the whole government.
- The goal is to support the creation of secure, responsive systems that keep up with evolving citizen and business needs.

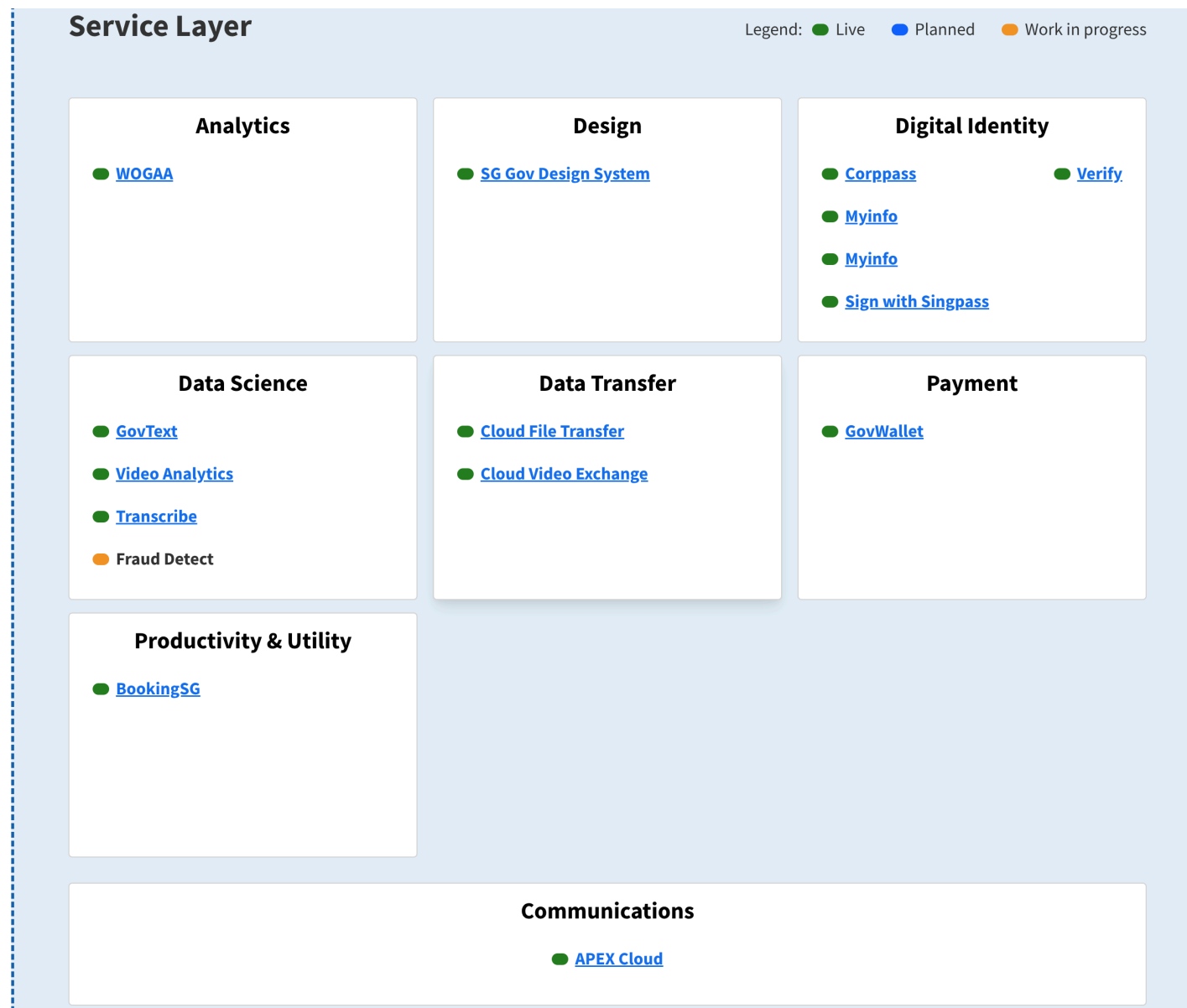
Architecture



Architecture - base layer



Architecture - service layer

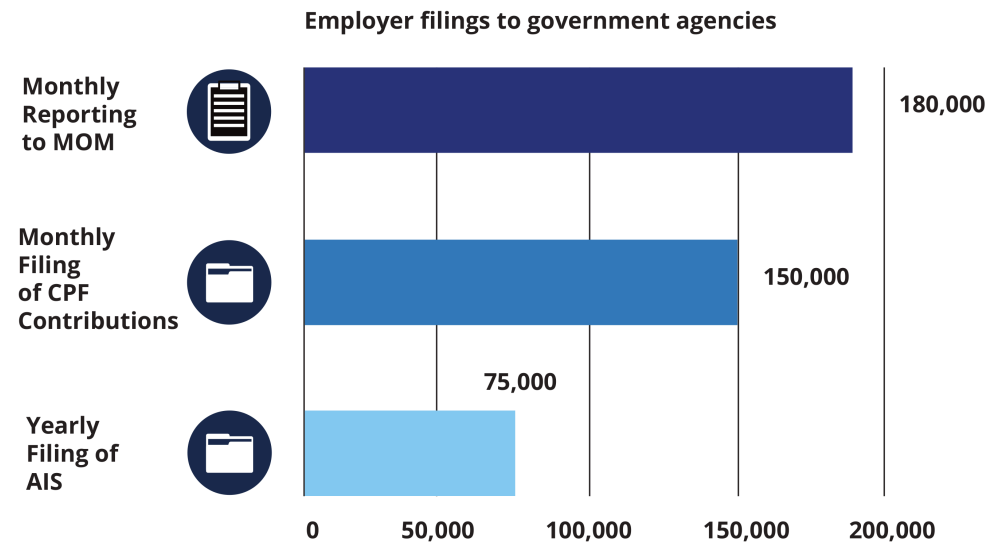


Success story

Streamlining Payroll Submissions in Singapore: APEX's
Solution for Efficient Government Reporting

The Problem

- Singapore-registered companies must regularly submit employee salary, workforce, and CPF contribution data to various government agencies (IRAS, MOM, CPFB).
- All required data already exists within companies' payroll or HR systems.
- The challenge is to integrate government submissions with these systems to streamline processes, reduce manual work, and save employers time and resources.



The solution

- APEX unifies APIs from multiple government agencies on one secure platform.
- It enables developers, including HR and payroll software providers, to integrate easily with these services.
- APEX removes the need for manual filing reducing administrative burden and supports better compliance with payroll-related regulations.

